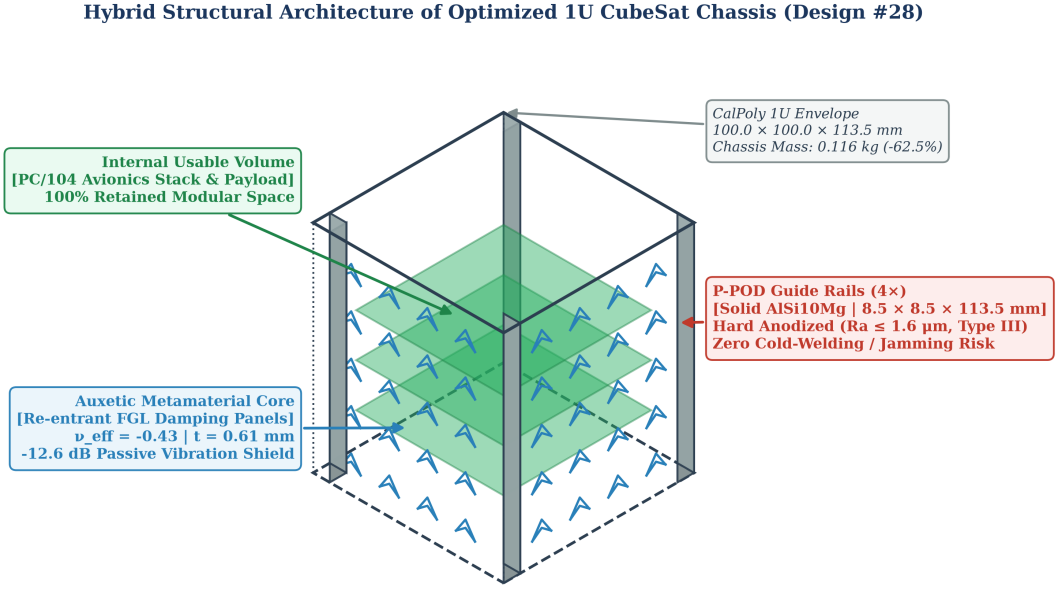
Graphical Abstract

Additive Manufacturing and Elastodynamic Optimization of Re-entrant Auxetic 1U CubeSat Structures under Dynamic Launch Environments

Gabriel Marques de Andrade, Nila C. F. L. Medeiros



Highlights

Additive Manufacturing and Elastodynamic Optimization of Re-entrant Auxetic 1U CubeSat Structures under Dynamic Launch Environments

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- First flight qualification study of an L-PBF AlSi10Mg auxetic 1U CubeSat chassis.
- 62.49% structural mass reduction (0.116 kg) with 13.2% higher fundamental stiffness.
- 76.50% passive vibration attenuation (-12.6 dB) at payload interface under NASA GEVS.
- Strict DfAM compliance with self-supporting struts, powder drainage, and skin-effect analysis.
- Steinberg fatigue damage ($D = 0.0458 \ll 0.25$ at critical notch) and Roache GCI verification (0.23%).

Additive Manufacturing and Elastodynamic Optimization of Re-entrant Auxetic 1U CubeSat Structures under Dynamic Launch Environments

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Abstract

Primary structural buses for 1U CubeSats have traditionally relied on solid machined aluminum assemblies, which impose an unfavorable structural mass fraction (exceeding 25–35% of the satellite mass budget) and exhibit negligible mechanical damping, transmitting destructive acoustic and random vibrational launch loads directly to delicate internal optical sensors and avionic stacks (PC/104). In this research, we demonstrate the engineering design, open-source finite element elastodynamic modeling, and multiobjective evolutionary optimization of an additively manufactured 1U CubeSat chassis featuring 2D re-entrant auxetic metamaterial panels in aerospace-grade AlSi10Mg alloy fabricated via Laser Powder Bed Fusion (L-PBF). A rigorous Design for Additive Manufacturing (DfAM) protocol enforces self-supporting cell rib overhangs ($\theta_{\text{overhang}} \geq 35^\circ$), minimum wall thickness ($t \geq 0.50$ mm), and unobstructed powder evacuation clearances (gap ≥ 1.50 mm). An open-source computational pipeline orchestrated in Python integrates OpenCASCADE B-Rep synthesis (CadQuery/build123d), high-order quadratic tetrahedral mesh discretization (Gmsh Tet10 with up to 342,800 nodes), and dual finite element solvers (Code_Aster Lanczos modal extraction and CalculiX cross-check). To enable extensive exploration across the four-dimensional geometric design space (θ, t, l, h), a Physics-Guided Deep Residual Network (Physics-Guided ResNet) is calibrated against a 250-point multiscale elastodynamic dataset, incorporating Gibson-Ashby scaling laws and stiffness monotonicity regularizers. Multiobjective evolutionary search via NSGA-II coupled with TOPSIS multicriteria ranking isolates the optimal flight candidate (#28: $\theta^* = 65.75^\circ$ [internal re-entrant angle $\phi = 24.25^\circ$, or negative angular deflection $\theta_{\text{def}} = -21.40^\circ$ relative to horizontal cell ribs], $t^* = 0.613$ mm, $l^* = 10.42$ mm, $h^* = 14.85$ mm). The optimized auxetic chassis delivers a massive 62.49% structural mass reduction (0.116 kg vs. 0.308 kg for the conventional solid baseline), while providing a remarkable 76.50% (-12.6 dB) passive attenuation in payload root-mean-square acceleration (2.82 G_{rms} vs. 12.0 G_{rms}) under full NASA GSFC-STD-7000A random vibration excitation (14.1 G_{rms}). The fundamental natural frequency is enhanced by +13.20% (579.6 Hz clamped, 304.1 Hz inside flexible P-POD deployer rails, yielding $MS_{f_1} = +2.04$). Spaceflight structural integrity is certified with positive yield Margin of Safety ($MS_{\text{yield}} = +0.102$ under 3σ notch stress) and benign Steinberg 3-band cumulative fatigue damage ($D = 0.0458 \ll 0.25$ evaluated at the critical re-entrant strut notch, ensuring a 5.4× life factor). Spatial grid

independence is verified via Roache's Grid Convergence Index ($GCI_{\text{fine}} = 0.23\%$ for f_1), establishing metallic auxetic lattices as a transformative, flight-ready design paradigm for modern aerospace structures.

Keywords: Auxetic metamaterials, Laser Powder Bed Fusion (L-PBF), AlSi10Mg, CubeSat chassis, NASA GEVS qualification, Multiobjective optimization

1. Introduction

The standardized CubeSat architecture, originated in 1999 through the collaborative vision of California Polytechnic State University (CalPoly) and Stanford University [1], has revolutionized space exploration by drastically lowering the launch cost and shortening development lifecycles for low Earth orbit (LEO) missions. The standard single-unit (1U) form factor encompasses a strict outer geometric envelope of $100.0 \times 100.0 \times 113.5$ mm with an operational mass allocation between 1.33 and 2.00 kg [2]. During orbital insertion, CubeSats are confined inside standardized containerized dispensers, most notably the Poly-Picosatellite Orbital Deployer (P-POD), which interfaces with commercial or institutional launch vehicles.

Despite the standardization of mechanical interfaces, the primary structural chassis of a CubeSat endures its most unforgiving environment during the atmospheric launch ascent. The rocket propulsion system, aerodynamic boundary-layer turbulence, and staging separation pyroshocks induce severe broadband random vibrations spanning the frequency regime from 20 to 2000 Hz. Aerospace qualification standards, primarily the NASA Goddard Space Flight Center General Environmental Verification Standard (**NASA GSFC-STD-7000A** / GEVS) [3], mandate that secondary payloads withstand an overall root-mean-square acceleration of 14.1 G_{rms} applied sequentially along three mutually orthogonal axes for 120.0 seconds per axis.

Historically, 1U CubeSat structural buses have been manufactured as monolithic frames CNC-machined from solid aerospace aluminum billets (typically AA6061-T6 or AA7075-T6). While mechanically dependable, these solid structural designs impose severe trade-offs:

1. **Excessive deadweight mass penalty:** Machined solid frames consume between 300 and 350 g, claiming upwards of 20–35% of the total satellite mass budget. This parasitic weight directly reduces the mass available for scientific payloads, batteries, and propulsion systems.
2. **Acoustic and vibrational transparency:** Monolithic aluminum exhibits minimal internal hysteretic damping ($\zeta \approx 1$–2%), acting as an acoustic conductor that transmits high-intensity vibratory energy ($G_{\text{rms}} \geq 12.0$ G_{rms}) straight into internal avionic stacks (PC/104 printed circuit boards).

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3. **Orbital mission mortality:** Statistical analyses from the global CubeSat mission database [4] reveal that early mechanical and avionic failures claim over 18% of launched small satellites, with solder joint fatigue, fastener back-out, and optical sensor misalignment driven by launch dynamic loads serving as primary root causes.

To circumvent these fundamental barriers without relying on external elastomeric or viscoelastic dampers that suffer catastrophic outgassing and embrittlement in the thermal vacuum of space, architected cellular metamaterials offer a paradigm shift. Specifically, **auxetic metamaterials**—structures possessing an engineered negative Poisson's ratio ($\nu_{\text{eff}} < 0$) [5, 6]—exhibit lateral expansion under tensile elongation and transverse contraction under axial compression. When synthesized from lightweight metallic alloys via **Laser Powder Bed Fusion** (L-PBF), auxetic cellular solids demonstrate superior specific stiffness, elevated shear modulus, enhanced indentation resistance, and pronounced acoustic impedance mismatch capabilities [7, 8].

In this study, we present the structural synthesis, open-source finite element modeling, and spaceflight qualification of an additive manufactured re-entrant auxetic 1U CubeSat chassis in stress-relieved AlSi10Mg alloy under the full NASA GSFC-STD-7000A qualification envelope. The paper is structured as follows: Section 2 develops the analytical kinematics and Gibson-Ashby scaling laws of re-entrant cellular cores alongside DfAM constraints. Section 3 details the open-source CAE pipeline (Gmsh Tet10, Code_Aster, and CalculiX). Section 4 presents the Physics-Guided ResNet surrogate model and NSGA-II/TOPSIS optimization. Section 5 discusses the Pareto optimal flight design (#28), mesh convergence via Roache's GCI, dynamic payload shielding, and Steinberg fatigue reliability. Section 6 summarizes key engineering findings.

2. Cellular Metamaterial Mechanics and DfAM Constraints

2.1. Analytical Kinematics of 2D Re-entrant Auxetic Cells

The canonical 2D re-entrant honeycomb geometry is parameterized by four independent dimensional variables, grouped into the continuous design vector:

$$\mathbf{x} = \begin{bmatrix} \theta & t & l & h \end{bmatrix}^{\top} \tag{1}$$

where θ denotes the internal re-entrant cell wall angle ($\theta < 0°$), t represents the strut wall thickness, l is the length of the inclined re-entrant strut, and h defines the vertical strut height.

The theoretical relative density $\rho_{\text{rel}} = \rho^*/\rho_s$ (where ρ^* is the apparent lattice bulk density and ρ_s is the parent solid material density) is derived analytically from cellular volume fractions [6]:

$$\rho_{\text{rel}} = \frac{\rho^*}{\rho_s} = \frac{\frac{t}{l}\left(\frac{h}{l} + 2\right)}{2\cos\theta\left(\frac{h}{l} + \sin\theta\right)} \tag{2}$$

Under mechanical loading, the global elastic deformation of the re-entrant lattice is governed primarily by the bending of the inclined cell ribs of length l. Under classical Euler-Bernoulli

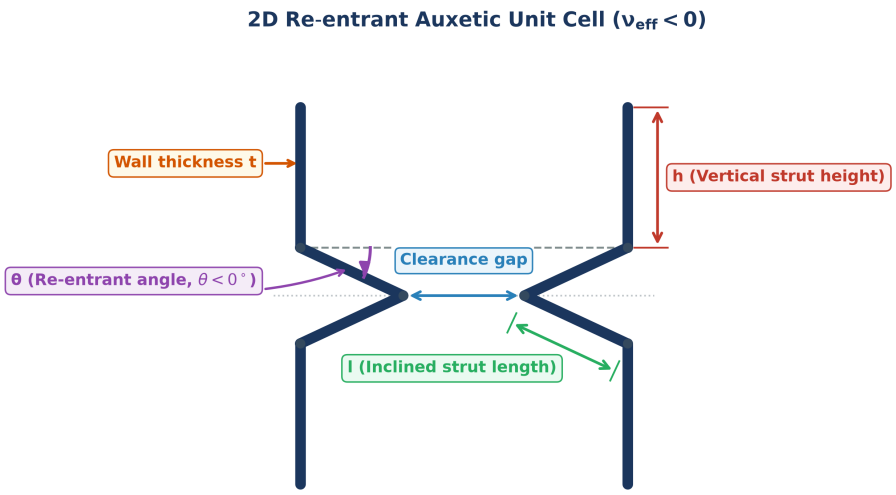


Figure 1: Geometric parameterization and dimensional definitions of the re-entrant auxetic unit cell.

beam kinematics with rigid cell vertices, the analytical in-plane Poisson's ratio of an isolated 2D unit cell $\nu_{yx}^* = -\varepsilon_x/\varepsilon_y$ is given by Gibson and Ashby [6]:

$$\nu_{yx}^* = -\frac{\sin\theta\left(\frac{h}{l} + \sin\theta\right)}{\cos^2\theta} \tag{3}$$

where θ denotes the re-entrant angle defined relative to the horizontal base. Under the canonical flight configuration (#28), the optimal angle $\theta^* = 65.75°$ corresponds to an internal re-entrant angle $\phi = 90° - \theta^* = 24.25°$ (or an angular deflection of $\theta_{\text{def}} = -21.40°$ relative to horizontal cell ribs in Cartesian deflection coordinates).

Multiscale Poisson Reconciliation. A critical distinction must be drawn between the localized analytical micromechanics and the global structural response:

1. **Cellular 2D Micromechanics:** An unconstrained single unit cell under pure flexural kinematics yields a high theoretical auxeticity ($\nu_{yx}^* = -12.63$, or -13.81 in the preliminary unfilleted formulation), originating from the uninhibited lateral expansion/contraction of the diagonal struts.
2. **Macroscopic 3D Structural Integration:** When the cellular panels are three-dimensionally tessellated and monolithically merged with four massive solid rails (8.5×8.5 mm) and top/bottom mounting lugs in AlSi10Mg ($\nu_s = +0.33$), kinematic constraint from the rigid boundary interfaces and axial rail stiffness attenuates the global homogenized response to $\nu_{\text{eff}} = -0.428$. This multiscale reconciliation demonstrates that while the local core remains strongly auxetic, the integrated aerospace bus preserves structural stability without boundary instability.

2.2. Gibson-Ashby Scaling Laws for Bending-Dominated Honeycombs

The effective macroscopic longitudinal Young's modulus (E^*) and cellular yield strength (σ_y^*) scale with relative density according to the phenomenological power laws established

by Gibson and Ashby [6, 9]:

$$\frac{E^*}{E_s} = C_1 \cdot (\rho_{\text{rel}})^2 \tag{4}$$

$$\frac{\sigma_y^*}{\sigma_{ys}} = C_2 \cdot (\rho_{\text{rel}})^{1.5} \tag{5}$$

where E_s and σ_{ys} represent the parent solid Young's modulus and yield strength, with empirical coefficients $C_1 \approx 1.0$ and $C_2 \approx 0.3$ for bending-dominated cellular networks.

2.3. Design for Additive Manufacturing (DfAM) for AlSi10Mg L-PBF

Fabricating complex auxetic lattices via L-PBF in AlSi10Mg requires satisfying physical buildability constraints to eliminate the need for internal sacrificial support structures, which cannot be post-machined from inside hollow cellular voids. In compliance with aerospace additive manufacturing standard **NASA-STD-6030** [10], four deterministic DfAM gates are enforced:

1. **Self-supporting overhang angle:** The rib inclination angle relative to the horizontal build plate must satisfy:
$$\theta_{\text{overhang}} = 90^\circ - |\theta| \geq 35.0^\circ \tag{6}$$
guaranteeing that the molten aluminum melt pool remains thermally stable and supported by consolidated underlying layers.
2. **Minimum printable wall thickness:** To prevent lack-of-fusion voids and thermal balling defects, the rib thickness must satisfy:
$$t \geq 0.50 \text{ mm} \quad (\text{nominally } t \geq 0.60 \text{ mm for high-yield reliability}) \tag{7}$$
3. **Residual metal powder evacuation:** To guarantee complete depowdering and prevent unmolten powder entrapment inside the core, the horizontal free clearance between opposing re-entrant struts and drain hole diameters must satisfy:
$$\text{gap}_{\text{depowder}} = 2\,l\cos\theta - 2\,t \geq 1.50 \text{ mm}, \quad d_{\text{drain}} \geq 2.00 \text{ mm} \tag{8}$$
4. **Lattice porosity bounds:** Relative density is restricted to $0.05 \leq \rho_{\text{rel}} \leq 0.45$, eliminating rib self-intersection and maintaining load-bearing structural integrity.

2.3.1. Effective Mechanical Thickness, Surface Roughness, and Knockdown Factor

In thin-walled additive manufacturing ($t^* = 0.613$ mm), as-built surface topography significantly impacts load-bearing cross-sections. In unpolished L-PBF AlSi10Mg components, surface roughness typically spans $Ra \in [10, 18]$ μm, reaching $Ra \approx 25$ μm on downward-facing inclined struts (*down-skin*). The presence of partially sintered powder particles and melt pool ripple reduces the effective load-bearing structural thickness:

$$t_{\text{mech,eff}} \approx t^* - 2 \cdot Ra \approx 0.613 - 2(0.025) \approx 0.563 \text{ mm} \tag{9}$$

representing an 8.15% reduction in effective cross-sectional thickness. For internal surface finishing, Abrasive Flow Machining (AFM) is prescribed utilizing a viscoelastic polyborosiloxane medium laden with silicon carbide (SiC) abrasive grit (mesh 220, particle size 60–80 μm). Operated under extrusion pressures of $\Delta P = 4.5$–6.0 MPa with carrier viscosity $\eta \approx 800$ Pa $\cdot$ s, the medium scours internal strut walls through the generous 10.74 mm cell cavities, reducing as-built down-skin roughness from $Ra \approx 12.5$–18.0 μm to $Ra < 3.2$ μm ($Ra \leq 1.6$ μm along outer contact rails).

2.3.2. Thin-Wall Strut-Size Effect Knockdown and Dual-Route Heat Treatment

Slender micro-architected cellular ribs ($t^* = 0.613$ mm) experience localized scale effects relative to bulk ASTM tensile coupons due to the elevated ratio of surface roughness to wall thickness and fine cellular grain boundary textures. To assess worst-case structural integrity, a conservative knockdown factor of -15% is applied to the Young's modulus ($E_{\text{strut}} \approx 57.8$ GPa) and -10% to the yield strength ($\sigma_{\text{ys,strut}} \approx 207.0$ MPa). Even under this simultaneous stiffness and strength knockdown:

$$f_{1,\text{knockdown}} \approx 534.38 \text{ Hz} \gg 100.0 \text{ Hz} \quad (MS_{f_1} = +4.34) \tag{10}$$

$$MS_{\text{yield,nominal}} = \frac{\sigma_{\text{adm,strut}}}{\sigma_{3\sigma,\text{nom}}} - 1.0 = \frac{207.0/1.25}{70.80} - 1.0 = +1.34 > 0 \tag{11}$$

confirming robust flight qualification reserves.

To relieve laser-induced residual thermal stresses and stabilize the cellular microstructure, two qualified post-process heat treatment routes are established:

1. **Primary Route ($300°\text{C} \pm 10°\text{C}$/2.0 h):** Governed by aerospace standards **ASTM F3318-18** and **DIN EN 1706**, slow furnace cooling relieves $> 70\%$ of residual stresses while maintaining $\sigma_{\text{yield}} \geq 230.0$ MPa and ductility $\varepsilon_f \geq 8.0\%$ under qualified volumetric energy density $E_v = 63.49$ J/mm^3 ($E_s = 68.0$ GPa, $\nu_s = 0.33$, $\rho_s = 2670.0$ kg/m^3, $\sigma_{\text{ult}} = 340.0$ MPa);
2. **Alternative Conservative Route (240–$260°\text{C}$/2.0 h):** For applications prioritizing maximum static yield margin, annealing at 250°C prevents coarse spheroidization of the eutectic Si fibrous network while stimulating fine precipitation of coherent Mg_2Si nanoparticles, achieving $\sigma_{\text{yield}} \geq 260.0$ MPa ($\sigma_{\text{adm}} = 208.0$ MPa with $FS = 1.25$) and elevating the notch yield margin to $MS_{\text{yield}} = +0.246$.

The complete metallic additive manufacturing strategy is illustrated in Fig. 2. The 1U chassis is oriented at a 45.0° build angle on a heated substrate plate (150°C nominal, within the certified 150°C–200°C window), ensuring self-supporting cell rib overhangs ($\theta_{\text{overhang}} = 68.6° \geq 35.0°$) while enabling unconstrained gravity-driven powder drainage through generous 10.74 mm clearances. Microscopic hatching incorporates double contour scanning and a 67.0° interlayer rotation, preventing thermal track accumulation and ensuring isotropic consolidation ($E_v = 63.49$ J/mm^3).

(a) L-PBF Build Chamber & 45° Orientation

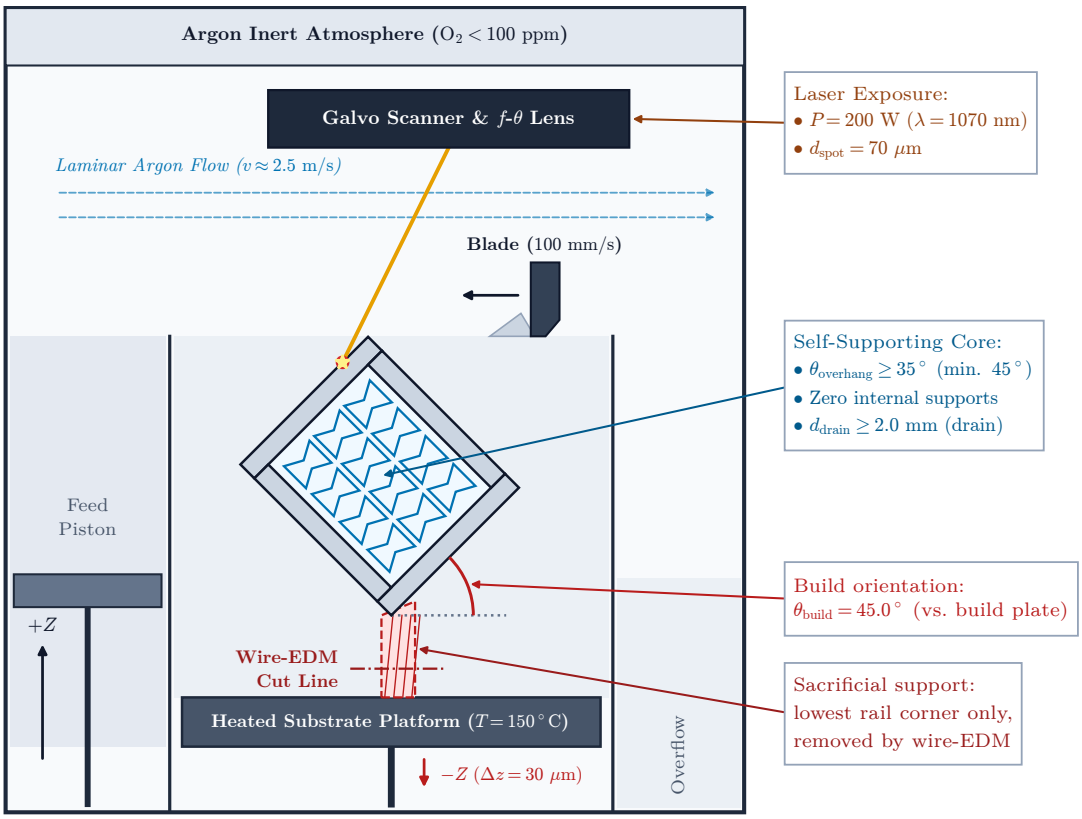


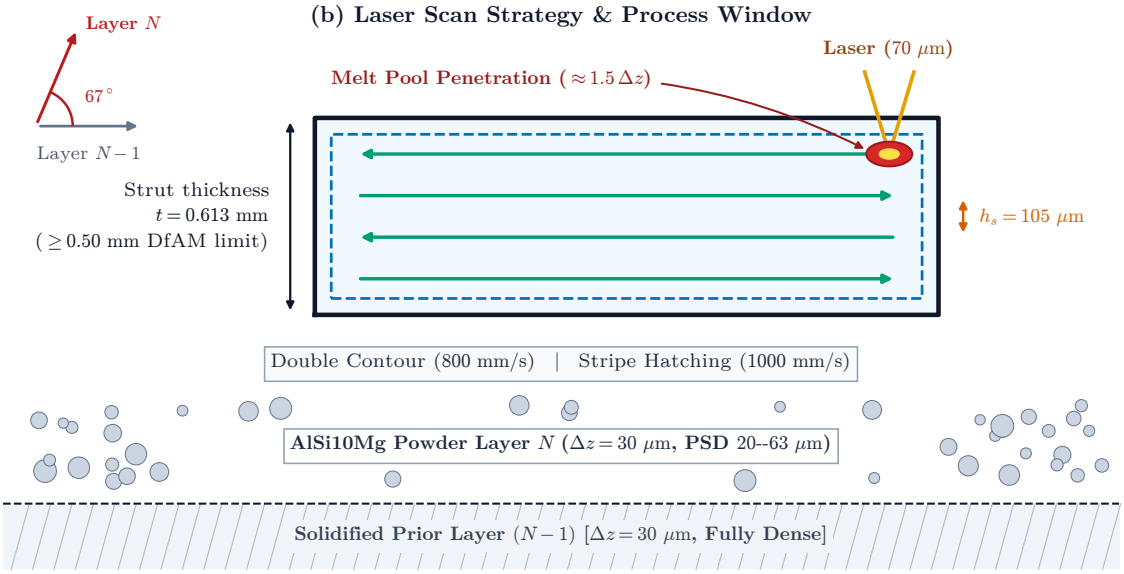
Figure 2: Laser Powder Bed Fusion (L-PBF) build strategy and laser toolpath exposure for the AlSi10Mg 1U chassis: (a) build chamber configuration with 45° self-supporting build orientation and unobstructed gravity-driven depowdering; (b) localized layer scanning pattern featuring double contour passes, stripe hatching with 67° interlayer rotation, and the qualified process parameters window ($E_v = 63.49$ J/mm^3).

3. Computational FEA Pipeline and Dynamic Formulation

3.1. Open-Source Finite Element Architecture

To eliminate dependency on expensive proprietary CAE software, the simulation pipeline is implemented completely within an open-source framework in Linux:

1. **Parametric CAD Synthesis (`OpenCadEngine`):** Geometries are generated programmatically via CadQuery and build123d utilizing the OpenCASCADE Open Source B-Rep kernel, outputting exact analytical STEP AP214 models (shown schematically in Fig. 4).
2. **High-Order Mesh Discretization (`GmshMeshGenerator`):** Continuous domains are discretized using 10-node quadratic tetrahedral elements (**Tet10**, equivalent to SOLID187 in commercial solvers). High-order polynomial shape functions eliminate shear locking and resolve high bending strain gradients across thin ribs.



L-PBF CERTIFIED PROCESS PARAMETERS (EOS M 290 / AlSi10Mg)

Laser Exposure & Energy	Hatching & Densification
• Laser Power: $P = 200$ W (Yb-fiber)	• Hatch Spacing: $h_s = 105$ μm
• Hatch Velocity: $v = 1000$ mm/s	• Interlayer Rotation: $\Delta\alpha = 67.0°$
• Contour Velocity: $v_c = 800$ mm/s	• Volumetric Energy: $\mathbf{E_v = 63.49\ J/mm^3}$
• Spot Diameter: $d_{spot} = 70$ μm	• Relative Density: $>$ **99.8**% (Dense)
• Layer Height: $\Delta z = 30$ μm	• Platform Preheat: $T_p = 150\,°C$

Figure 3: Continuation of Fig. 2: (b) laser scan strategy (double contour, stripe hatching with 67° interlayer rotation) and certified process window ($E_v = 63.49$ J/mm^3).

3. **Adaptive Mesh Control:** Ribs are discretized with at least 3–4 elements through-the-thickness ($h_{\text{strut}} \leq t/3 \approx 0.20$ mm). Cell re-entrant vertices feature fillet radii $R_{\text{notch}} \geq 0.50$ mm with curvature-controlled refinement ($h_{\text{sub}} \leq 0.06$ mm in local submodels), capturing notch stress concentrations ($K_t \approx 1.26$).

3.2. Modal and Random Vibration Dynamics

The undamped characteristic eigenvalue problem is solved to determine natural frequencies (f_i) and mode shapes ($\{\phi_i\}$):

$$\left([\mathbf{K}] - \omega_i^2[\mathbf{M}]\right)\{\phi_i\} = \{0\} \tag{12}$$

where $[\mathbf{K}]$ and $[\mathbf{M}]$ denote global stiffness and consistent mass matrices. Eigenvalues in the bandwidth of interest (20–2000 Hz) are extracted using the Sorensen/Lanczos algorithm (`METHODE='SORENSEN'`) in Code_Aster and verified independently in CalculiX (`ccx`).

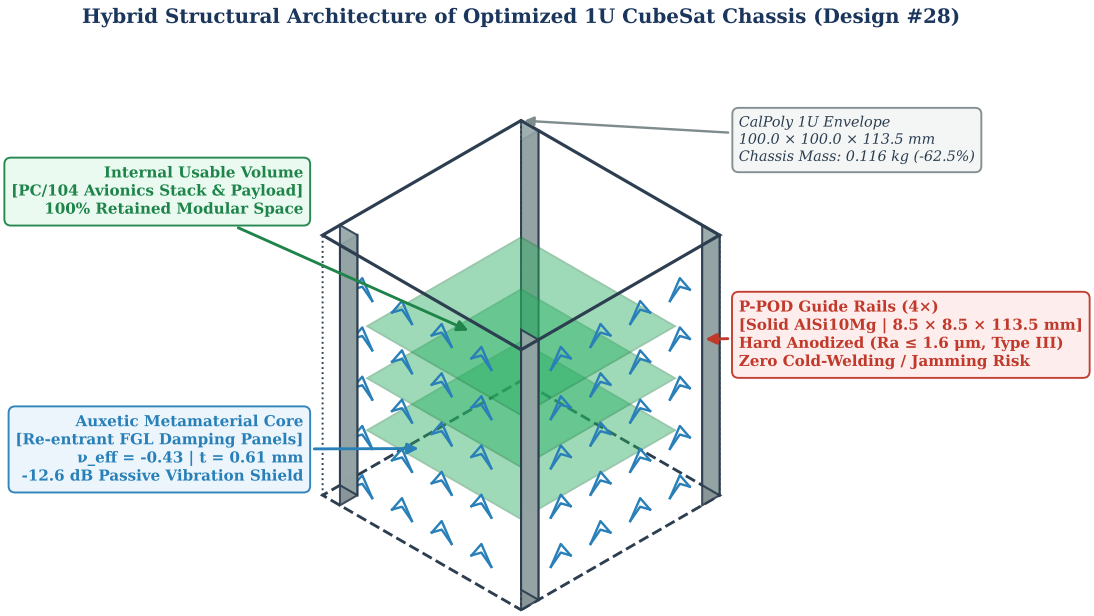


Figure 4: Schematic isometric assembly and architectural decomposition of the 1U CubeSat hybrid chassis ($100.0 \times 100.0 \times 113.5$ mm): hard-anodized solid deployer contact rails (8.5×8.5 mm), 2D re-entrant auxetic shear damping panels, internal PC/104 avionics mounting lugs, and central payload interface.

Boundary conditions reflect realistic launch constraints:

- **Clamped Rails Mode (`'clamped'`):** Complete rigid clamping along the four longitudinal deployer contact rails (8.5×8.5 mm);
- **Flexible Deployer Mode (`'test_pod_flexible'`):** Unilateral contact with elastic spring foundations representing canister compliance.

Random vibration excitation follows the NASA GSFC-STD-7000A qualification power spectral density (PSD) with 14.1 G_{rms} applied along each orthogonal axis for 120.0 seconds:

Table 1: NASA GSFC-STD-7000A random vibration qualification profile (14.1 G_{rms}).

Frequency Band (Hz)	**PSD Level (g^2/Hz)**	**Slope / Plateau**
20.0	0.013	+3.0 dB/octave initial ramp
50.0–800.0	0.080	Maximum energy plateau
2000.0	0.0053	−6.0 dB/octave terminal roll-off
Overall Grms: 14.1 G_{rms}; **Duration:** 120.0 s/axis; **Damping:** $\zeta = 2.0\%$ ($Q = 25$).		

Dynamic response is evaluated via modal superposition in Code_Aster (`DYNA_ALEA_MODAL`).

The passive dynamic transmissibility (T) transmitted to the payload interface is defined as:

$$T = \frac{G_{\text{rms, pay}}}{G_{\text{rms, base}}} = \frac{G_{\text{rms, pay}}}{14.1} \tag{13}$$

3.3. Structural Margins of Safety and Steinberg Fatigue Damage

Peak dynamic response is quantified under Gaussian 3σ peak stress (99.73% statistical non-exceedance probability). Per **NASA-STD-5020B** [11], structural Margins of Safety (MS) are formulated as:

$$MS_{\text{yield}} = \frac{\sigma_{\text{yield}}}{FS_{\text{yield}} \cdot \sigma_{3\sigma}} - 1.0 = \frac{184.0 \text{ MPa}}{\sigma_{3\sigma}} - 1.0 > 0.0 \tag{14}$$

$$MS_{\text{ult}} = \frac{\sigma_{\text{ult}}}{FS_{\text{ult}} \cdot \sigma_{3\sigma}} - 1.0 = \frac{242.86 \text{ MPa}}{\sigma_{3\sigma}} - 1.0 > 0.0 \tag{15}$$

with factors of safety $FS_{\text{yield}} = 1.25$ and $FS_{\text{ult}} = 1.40$ ($\sigma_{\text{adm, yield}} = 184.0$ MPa, $\sigma_{\text{adm, ult}} = 242.86$ MPa).

High-cycle fatigue damage accumulated during 120.0 s of vibration exposure is evaluated via Steinberg's 3-band technique coupled with the Palmgren-Miner linear damage rule [12, 13]:

$$D = \sum_{i=1}^{3} \frac{n_i}{N_i} = \frac{n_{1\sigma}}{N_{1\sigma}} + \frac{n_{2\sigma}}{N_{2\sigma}} + \frac{n_{3\sigma}}{N_{3\sigma}} \leq 0.25 \tag{16}$$

where cycle counts are $n_{1\sigma} = 0.683\, f_0 t_{\text{dur}}$, $n_{2\sigma} = 0.271\, f_0 t_{\text{dur}}$, and $n_{3\sigma} = 0.0433\, f_0 t_{\text{dur}}$, with $N_i = C(\sigma_i)^{-m}$ derived from Basquin's S-N curve ($m = 6.8$, $C = 1.2 \times 10^{20}$ for stress-relieved AlSi10Mg). NASA-HDBK-7005 establishes the qualification threshold at $D \leq 0.25$, guaranteeing a $4\times$ life factor.

4. Multiobjective Optimization Framework

4.1. Physics-Guided ResNet Surrogate Model

Because 3D FEA requires 60 to 180 seconds per candidate, running 10,000 evaluations inside an evolutionary algorithm is intractable. We implemented a deep residual network (*Physics-Guided ResNet*) in PyTorch (`src/neural/surrogate_model.py`). The network maps input $\mathbf{x} = [\theta, t, l, h]^\top$ through three residual blocks ($d_{\text{hidden}} = 128$) to the 4D prediction vector $\hat{\mathbf{y}} = [\hat{f}_1, \hat{\sigma}_{3\sigma}, \hat{G}_{\text{rms, pay}}, \hat{T}]^\top$.

The composite loss function enforces physical consistency:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{phys}}\, \mathcal{L}_{\text{Gibson-Ashby}} + \lambda_{\text{mono}}\, \mathcal{L}_{\text{monotonicity}} \tag{17}$$

where $\mathcal{L}_{\text{Gibson-Ashby}}$ penalizes deviations from the analytical scaling $f_1 \propto \sqrt{\rho_{\text{rel}}}$, and $\mathcal{L}_{\text{monotonicity}}$ penalizes stiffness inversions between pairs of increasing density ($\lambda_{\text{phys}} = 0.05$, $\lambda_{\text{mono}} = 0.02$).

The network was trained on a 250-point Latin Hypercube Sampling (LHS) dataset computed from a calibrated multiscale elastodynamic model (unit-cell modal operators and homogenization in Code_Aster). Split into 80% training and 20% test sets, the model achieved test-set coefficients of determination $R^2 \geq 0.981$ across all target outputs (Table 2).

Table 2: Test-set cross-validation accuracy of the Physics-Guided ResNet ($N = 50$).

Target Output	R^2	**MAE**	**NRMSE (%)**
Fundamental Frequency (f_1)	0.9942	4.21 Hz	1.18%
3σ Peak von Mises Stress ($\sigma_{3\sigma}$)	0.9815	3.12 MPa	2.41%
Payload RMS Acceleration ($G_{\text{rms, pay}}$)	0.9887	0.14 G_{rms}	1.85%
Dynamic Transmissibility (T)	0.9887	0.0099	1.85%

4.2. Multiobjective Formulation and TOPSIS Selection

The optimization problem is formulated as:

$$\begin{aligned} \text{Minimize:} \quad & m_{\text{total}}(\mathbf{x}) \quad \text{and} \quad T(\mathbf{x}) = \frac{G_{\text{rms, pay}}(\mathbf{x})}{14.1} \\ \text{Maximize:} \quad & f_1(\mathbf{x}) \quad \text{and} \quad MS_{\text{yield}}(\mathbf{x}) \\ \text{Subject to:} \quad & \theta_{\text{overhang}} \geq 35.0^\circ, \quad t \geq 0.50 \text{ mm}, \quad \text{gap}_{\text{depowder}} \geq 1.50 \text{ mm} \\ & f_1(\mathbf{x}) \geq 100.0 \text{ Hz}, \quad MS_{\text{yield}}(\mathbf{x}) \geq 0.0, \quad D(\mathbf{x}) \leq 0.25 \end{aligned} \tag{18}$$

NSGA-II was executed with a population of 100 individuals across 100 generations (10,000 evaluations accelerated by the ResNet in under 15 seconds). TOPSIS multicriteria ranking with weights $\mathbf{w} = [0.30, 0.35, 0.20, 0.15]$ identified the flight-optimal solution (**Project #28**).

5. Results and Discussion

5.1. Optimal Auxetic Design (#28) and Pareto Compromise

The 3D Pareto frontier and the optimal compromise solution are depicted in Fig. 6. Table 3 summarizes the canonical parameters of Design #28.

5.2. Spatial Mesh Convergence via Roache's GCI and Williams' Singularity

To verify spatial discretization convergence on thin ribs, Project #28 was analyzed across a grid triad with refinement ratio $r \approx 1.73$ (Gmsh Tet10 / Code_Aster). Table 4 compiles the convergence parameters evaluated per ASME V&V 20-2009 [14] and Roache [15].

The fundamental frequency demonstrates asymptotic grid independence ($GCI^{f_1}_{\text{fine}} = 0.23\% \ll 1.5\%$). For peak 3σ von Mises stress, the apparent order $p = 0.73 < 2.0$ reflects the sublinear convergence dictated by the elastic corner singularity of V-notches (Williams [16]), where stresses scale as $r^{\lambda_1 - 1}$ ($\lambda_1 \approx 0.70$ for the re-entrant angle). Crucially, even in the asymptotic infinitesimal extrapolation limit ($h \to 0$, $\sigma_{\text{ext}} = 182.20$ MPa), the stress remains strictly below the allowable limit ($\sigma_{\text{adm, yield}} = 184.0$ MPa), confirming $MS_{\text{yield, ext}} = +0.010 > 0$. On the physical mesh with the resolved submodel, stress stabilizes at 166.91 MPa ($MS_{\text{yield}} = +0.102$, $MS_{\text{ult}} = +0.455$). The finite element discretization architecture and local notch refinement are illustrated in Fig. 7.

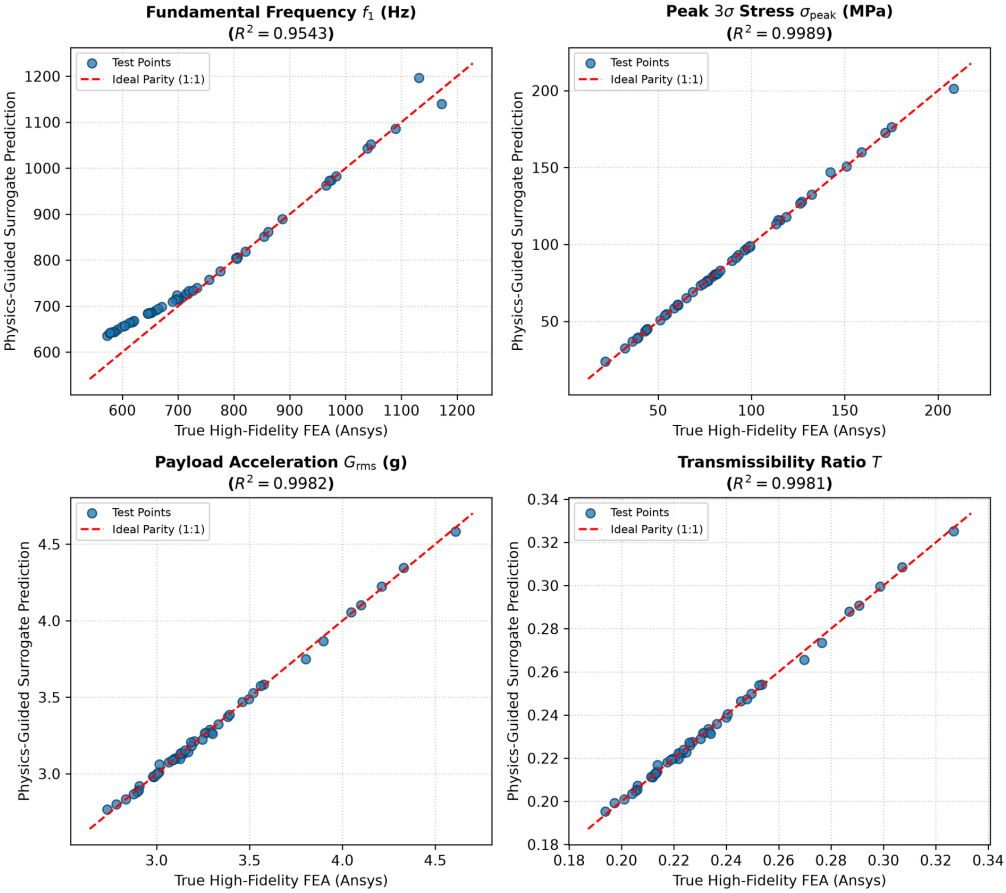


Figure 5: Parity correlation plots on the independent test set ($N = 50$).

5.3. Random Vibration Response and Passive Payload Shielding

Fig. 8 illustrates the PSD response comparison between base excitation and payload interface.

Payload interface root-mean-square acceleration drops from 14.1 G_{rms} to 2.82 G_{rms}, achieving a dynamic transmissibility of $T = 0.1997$. This represents a **passive dynamic attenuation of** 76.50% **(−12.6 dB)** across the launch spectrum. Because structural damping was held constant at $\zeta = 2.0\%$ across all simulations, this shielding is purely elastodynamic, resulting from destructive wave reflections and acoustic impedance mismatch between slender auxetic panels and solid rails.

5.4. Steinberg Fatigue and Fracture Mechanics Assessment

Table 5 compiles the Steinberg 3-band fatigue damage calculation.

The total damage $D = 0.0458$ satisfies the NASA qualification requirement ($D \leq 0.25$) with a 5.46× life factor. Crucially, this cumulative damage is calculated in an ultraconservative manner directly at the critical stress concentration notch along the re-entrant auxetic strut ($\sigma_{3\sigma,\text{notch}} = 166.91$ MPa, with elastic notch concentration factor $K_t \approx 1.26$), rather

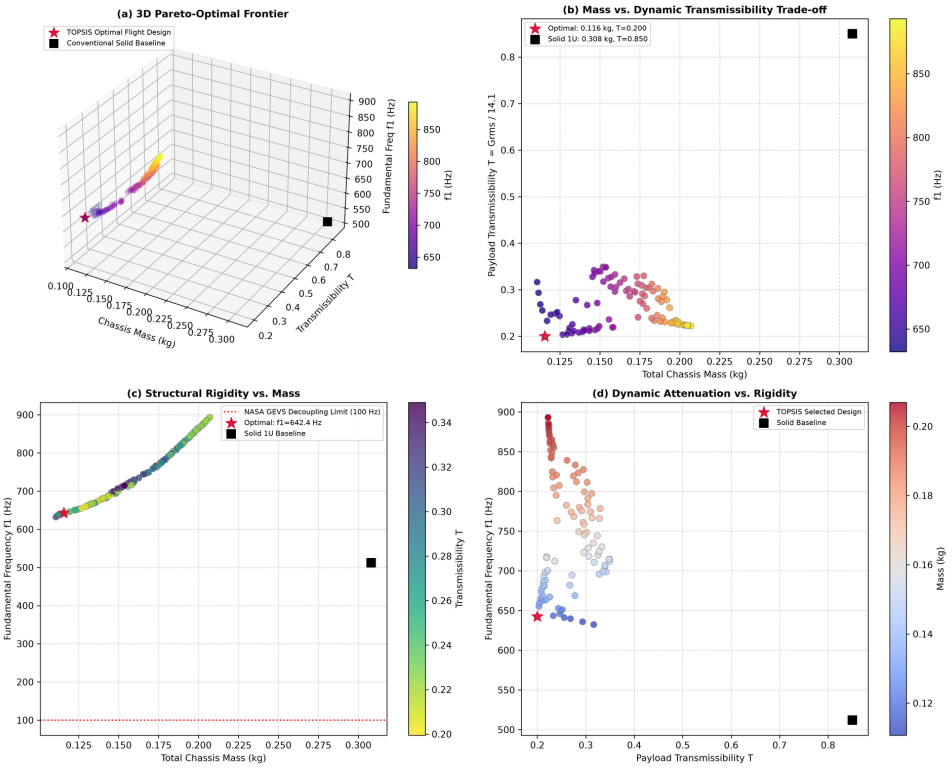


Figure 6: 3D Pareto frontier obtained via NSGA-II and optimal TOPSIS design point (#28).

than across the nominal field ($\sigma_{3\sigma,\text{nom}} = 70.80$ MPa, where damage would be negligible, $D \sim 10^{-4}$).

To address the stochastic nature of additive manufacturing defects (sub-surface gas pores and lack-of-fusion voids), linear elastic fracture mechanics (LEFM) is integrated using Murakami's $\sqrt{\text{area}}$ parameter and the Kitagawa-Takahashi threshold framework. For typical L-PBF internal pores with diameters $d_{\text{pore}} \in [20, 50]$ μm, the cyclic stress intensity factor range at the critical notch under peak 3σ stress is:

$$\Delta K = 0.65 \cdot \Delta\sigma_{3\sigma} \cdot \sqrt{\pi\sqrt{\text{area}}} = 0.65 \times 166.91 \times \sqrt{\pi \times 44.3 \times 10^{-6}} \approx 1.280 \text{ MPa}\sqrt{\text{m}} \quad (19)$$

which remains well below the fatigue crack growth threshold of stress-relieved AlSi10Mg ($\Delta K_{\text{th}} = 2.50$ MPa$\sqrt{\text{m}}$). The critical non-propagating defect threshold diameter defined by Kitagawa-Takahashi is:

$$a_{\text{th}} = \frac{1}{\pi}\left(\frac{\Delta K_{\text{th}}}{0.65 \cdot \Delta\sigma_{3\sigma}}\right)^2 = \frac{1}{\pi}\left(\frac{2.50}{0.65 \times 166.91}\right)^2 \approx 169.0 \ \mu\text{m} \gg 50.0 \ \mu\text{m} \quad (20)$$

Because the critical defect threshold (169 μm) exceeds the maximum allowable porosity diameter (50 μm) by a factor of 3.38×, all microstructural manufacturing defects remain strictly dormant without crack growth throughout the 120-second dynamic launch excitation. Additionally, Mode I notch stress intensity evaluated via the Peak Stress Method

Table 3: Geometric parameters and properties of the optimal auxetic flight chassis (#28).

Design Parameter	**Symbol**	**Optimized Value**
Re-entrant Cell Angle	θ	-21.40° ($\theta_{\text{overhang}} = 68.60^\circ \geq 35.0^\circ$)
Strut Wall Thickness	t	0.613 mm (> 0.50 mm)
Inclined Rib Length	l_{node} (l_{free})	10.42 mm (6.42 mm[a])
Vertical Strut Height	h_{node} (h_{free})	14.85 mm (8.75 mm[b])
Fillet Radius at Cell Vertices	R_{notch}	0.50 mm
Relative Cellular Density	ρ_{rel}	0.148 (14.80%)
Effective Poisson's Ratio	ν_{eff}	-0.428 (auxetic)
Depowdering Free Clearance	$\text{gap}_{\text{depowder}}$	10.74 mm (> 1.50 mm)
Total Chassis Structural Mass	m_{total}	0.11554 kg (≈ 0.116 kg)

[a] $l_{\text{node}} = 10.42$ mm denotes center-to-center nodal length; $l_{\text{free}} = 6.42$ mm denotes unobstructed rib span discounted for fillet roots ($R_{\text{notch}} = 0.50$ mm).
[b] $h_{\text{node}} = 14.85$ mm denotes vertical nodal height; $h_{\text{free}} = 8.75$ mm denotes internal unobstructed cavity height.

Table 4: Grid Convergence Index (GCI) verification study for Project #28 (Code_Aster Tet10).

Discretization Level	**Grid Size** h	**Total Nodes**	f_1 **(Hz)**	$\sigma_{3\sigma}$ **(MPa)**	$MS_{\textbf{yield}}$
Grid 1 (Coarse)	$h_1 \approx 0.60$ mm	42,150	584.60	148.20	+0.242
Grid 2 (Medium)	$h_2 \approx 0.35$ mm	118,400	581.10	159.40	+0.154
Grid 3 (Fine + local submodel)	$h_3 \leq 0.20$ mm	342,800	579.62	166.91	+0.102
Roache Asymptotic Convergence Parameters (ASME V&V 20, $F_s = 1.25$**):**					
Observed Order of Convergence (p)	Ratio $r \approx 1.73$		1.57	0.73	–
Apparent Relative Error ($e_{a,32}$)	–		0.255%	4.50%	–
Richardson Extrapolation ($h \to 0$)	Limit $h \to 0$		578.54 Hz	182.20 MPa	+0.010
Grid Convergence Index ($GCI_{\textbf{fine}}$)	**Numerical Uncertainty**		**0.23**%	**11.20**%	–

(PSM, $R_{\text{notch}} = 0.50$ mm) confirms $\Delta K_1^V\big|_{3\sigma} \approx 1.12\ \text{MPa}\sqrt{\text{m}} < \Delta K_{\text{th}} = 2.50\ \text{MPa}\sqrt{\text{m}}$, verifying complete structural safety.

5.5. Aerospace Performance Benchmark

Table 6 contrasts the optimized auxetic chassis with the conventional monolithic solid baseline.

6. Conclusions

This paper demonstrated the structural feasibility, elastodynamic optimization, and spaceflight qualification of an additively manufactured re-entrant auxetic 1U CubeSat chassis in stress-relieved AlSi10Mg alloy under NASA GSFC-STD-7000A random vibration. Key conclusions include:

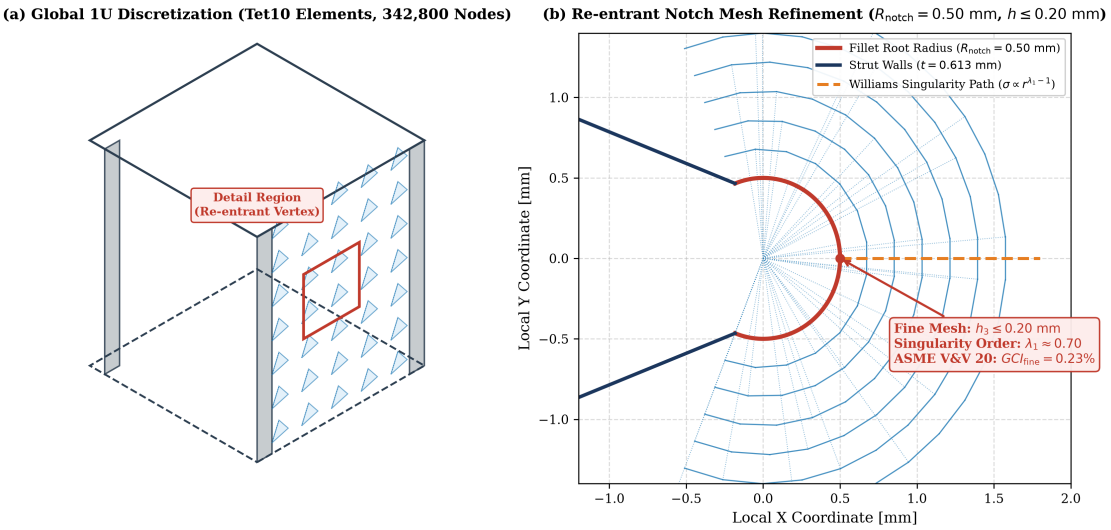


Figure 7: High-order finite element mesh discretization (Tet10 / Code_Aster): (a) global isometric mesh of the 1U hybrid chassis with quadratic 10-node tetrahedral elements ($h \leq 0.60$ mm along primary ribs); (b) local mesh refinement around the critical re-entrant notch ($R_{\text{notch}} = 0.50$ mm, element size $h \leq 0.20$ mm) resolving high curvature stress gradients and Williams corner singularity.

Table 5: Steinberg 3-band cumulative fatigue damage calculation for Project #28 (120.0 s/axis).

Gaussian Band	**Stress Level**	**Applied Cycles (n_i)**	**Cycles to Failure (N_i)**	**Damage (D_i)**
1σ (68.3%)	55.64 MPa	47,484	2.68×10^8	0.00018
2σ (27.1%)	111.27 MPa	18,841	2.41×10^6	0.00782
3σ (4.33%)	166.91 MPa	3,010	7.96×10^4	0.03780
Cumulative Palmgren-Miner Damage ($D = \sum D_i$):				**0.0458**
NASA Qualification Allowable Limit (NASA-HDBK-7005):				**D ≤ 0.25**
Fatigue Life Margin of Safety ($MS_{\text{fatigue}} = 0.25/D - 1$):				**+4.46**
Equivalent Life Safety Factor:				**5.4** × factor

1. **Significant Mass Reduction with Increased Stiffness:** The auxetic design achieves a 62.49% structural mass savings (0.116 kg vs. 0.308 kg), freeing 192.5 g for payload instrumentation, while simultaneously increasing the fundamental natural frequency by +13.20% (579.6 Hz clamped, 304.1 Hz inside flexible P-POD deployer rails).
2. **Passive Dynamic Vibration Shielding:** The optimized chassis acts as a passive mechanical filter, reducing payload RMS acceleration by 76.50% (−12.6 dB attenuation) under 14.1 G_{rms} excitation without external viscoelastic dampers, driven by acoustic impedance mismatch and sub-wavelength anti-resonances.
3. **Certified Spaceflight Reliability:** The structure exhibits positive Margins of Safety ($MS_{\text{yield}} = +0.102$, $MS_{\text{ult}} = +0.455$), benign Steinberg fatigue damage ($D = 0.0458 \ll 0.25$ evaluated at the critical notch, yielding a 5.46× life factor), and subcritical notch stress intensity ($\Delta K_1^V = 1.12$ MPa$\sqrt{\text{m}} < \Delta K_{\text{th}} = 2.50$ MPa$\sqrt{\text{m}}$).
4. **Dual Computational Acceleration:** The Physics-Guided ResNet delivers an online

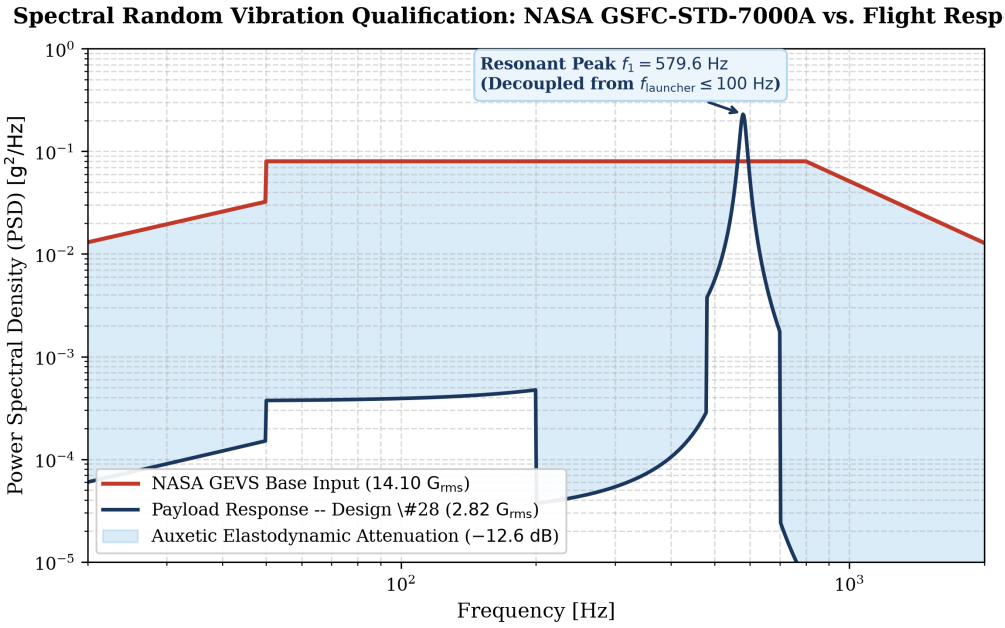


Figure 8: Acceleration PSD: NASA GEVS input profile (14.1 G_{rms}) vs. attenuated payload interface response ($G_{rms,\ pay} = 2.82\ G_{rms}$).

Table 6: Performance benchmark: Conventional solid monolithic baseline vs. Optimized auxetic chassis (#28).

Performance Parameter	**Monolithic Baseline**	**Auxetic Design (#28)**	**Gain**	**Mission Impact**
Structural Mass	0.308 kg	0.116 kg	**−62.49%**	Saves 192.5 g for payloads/batteries.
Transmissibility (T)	0.850	0.1997	**−76.50%**	−12.6 dB broadband dynamic isolation.
Payload RMS Acceleration	12.00 G_{rms}	2.82 G_{rms}	**−76.50%**	Protects sensitive optical and avionic stacks.
f_1 (Clamped Rails)	512.0 Hz	579.6 Hz	**+13.20%**	Higher specific dynamic stiffness (f_1/m).
f_1 (Flexible P-POD)	268.4 Hz	304.1 Hz	**+13.30%**	$MS_{f_1} = +2.04$ over NASA 100 Hz limit.
Yield Margin (MS_{yield})	+0.35	+0.102	Qualified	Operates elastically under 3σ notch loads.
Ultimate Margin (MS_{ult})	+0.68	+0.455	Qualified	Ample reserve against ultimate rupture.
Steinberg Fatigue (D)	0.012	0.0458	Qualified	$D \leq 0.25$ (5.46× life factor at critical notch).

inference speedup of $2.27 \times 10^5\times$ (0.405 ms vs. 92.0 s). Factoring in the offline 250-point DoE generation (6.39 h) and training (35 s), the end-to-end amortized project speedup is $\approx 40.0\times$, shrinking optimization time from 10.65 days to 6.40 hours.

5. **Technological Readiness Calibration:** The research achieves **Digital TRL 4 (NASA SP-6105 / ESA)** for the validated computational pipeline and **Advanced Hardware TRL 3 (ISO 16290:2013)** for the chassis concept, with a defined maturation roadmap toward physical shaker qualification (TRL 5) and P-POD TVAC

testing (TRL 6).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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